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Principle and antimicrobial efficacy of laser-activated irrigation: A narrative review

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Abstract

In the last two decades, the activation of root canal irrigants with pulsed lasers as an adjunct in root canal treatment has become increasingly popular. This narrative review explains the physical basics and the working mechanism of laser-activated irrigation (LAI), explores the parameters influencing LAI efficacy, considers historical evolutions in the field and summarizes laboratory and clinical evidence with emphasis on the antimicrobial action of LAI. Cavitation is the driving force behind LAI, with growing and imploding vapour bubbles around the laser tip causing various secondary phenomena in the irrigant, leading to intense liquid dynamics throughout the underlying root canal. High-speed imaging research has shown that laser wavelength, pulse energy, pulse length and fibre tip geometry are parameters that influence this cavitation process. Nevertheless, this has not resulted in standardized settings for LAI. Consequently, there is significant variability in studies assessing LAI efficacy, complicating the synthesis of results. Laboratory studies in extracted teeth suggest that, with regard to canal disinfection, LAI is superior to conventional irrigation and there is a trend of higher antimicrobial efficacy of LAI compared to ultrasonic activation. Clinical evidence is limited to trials demonstrating similar postoperative pain levels after LAI versus no activation or ultrasonic activation. Clinical evidence concerning the effect of LAI on healing of apical periodontitis as yet is scarce.

KEYWORDS

cavitation, Er:YAG, Er;Cr:YSGG, irrigation, laser, root canal disinfection

INTRODUCTION

One of the cornerstones in the treatment and prevention of apical periodontitis is the thorough cleaning and disinfection of the root canal system. It has been frequently demonstrated that traditional chemomechanical root canal preparation techniques, combining mechanical enlargement of the canal space and syringe-needle-based irrigation with antimicrobial and chelating agents, even when well performed, fail to consistently reach this goal (Nair et al., 2005). Although studies have demonstrated acceptable clinical success rates of primary root canal treatment (Ng et al., 2008), treatment failures do occur in about 1 out of 5 teeth with preoperative apical periodontitis. Two factors have been held responsible for this. First, there is the micro-anatomy of the root canal system, which can be extremely complex (Vertucci, 1984). Conventional canal preparation techniques using manual or mechanically driven root canal instruments in combination with traditionally (needle-syringe based) delivered

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irrigants are mainly addressing the main canals, and the cleaning of secondary anatomy like isthmuses, fins, cul-de-sacs, apical deltas and lateral canals is poor (Leoni et al., 2017; Siqueira Jr. et al., 1997). Second, there is the fact that in case of an established root canal infection, micro-organisms are not just floating in the canal space, but are organized as highly resistant microbial biofilms (Ricucci & Siqueira Jr., 2010). These are complex structures composed of multiple microbial species, embedded within a self-produced polymer matrix and attached to the root canal wall. The structural organization and inherent resistance of the intracanal biofilm makes it extremely hard to eradicate, especially within a complex micro-anatomy. Micro-organisms persisting in the canal are the most important risk factor for post-treatment apical periodontitis (Siqueira Jr. & Rocas, 2008). Therefore, there is a constant quest for methods to improve root canal cleaning and disinfection. One of the approaches is the activation of irrigants, aiming at improving dynamics, distribution and cleaning action of the root canal irrigating solution. Many methods of irrigant activation have been described previously (Gu et al., 2009), including the use of rotating or oscillating tips placed inside the canal, the use of pressure-alternating devices like multisonic irrigation (Zhang et al., 2019) or the used of pulsed lasers. The latter is referred to as laser-activated irrigation (LAI).

LASER-ACTIVATED IRRIGATION VERSUS CONVENTIONAL LASER ENDODONTICS

The first applications of lasers in root canal treatment involved the use of a laser in a canal after canal shaping and drying, in order to directly irradiate the root canal walls. To this end, a laser tip is placed in the apical area and withdrawn with a helicoidal motion in order to expose the root canal walls and any underlying secondary anatomy (like tubules, etc.) to the laser light, in order to promote cleaning and disinfection (Saunders et al., 1995). Modification of the root canal wall (like dentin melting, tubule occlusion, smear layer removal) has also been described as an objective (Levy, 1992). Typically, lasers in the visible or near-infrared spectrum are used for this, since these wavelengths are delivered through a glass fibre, which is both flexible and sufficiently small to allow introduction into most prepared root canals. Lasers in the mid-infrared region have also been tested for this purpose (Moritz et al., 1999). This approach is referred to as 'conventional laser endodontics'.

However, because this approach experienced difficulties with regard to uniform canal wall coverage (Yasuda et al., 2010), and disappointing cleaning and disinfection results (Hardee et al., 1994) due to poor antimicrobial action of involved wavelengths (Meire et al., 2012), the focus of attention shifted towards the indirect use of lasers in root canal disinfection. That is how LAI came into play. LAI is the activation of the root canal irrigant with pulsed lasers. In this case, after shaping, the canals are filled with irrigant, and pulsed laser irradiation is delivered into the irrigant, with the goal of promoting the irrigant's penetration into the canal system and enhancing its cleaning and disinfecting action.

WORKING MECHANISM OF LAI: THE POWER OF CAVITATION

The basic mechanism of action of LAI is the succession of very rapidly growing and imploding vapour bubbles within the irrigant. LAI is generated by erbium lasers (this shall be discussed further) working in pulsed mode. This means that these lasers do not emit continuously, but deliver short pulses of laser energy. The duration of each pulse is fairly short, in the order of microseconds (roughly varying between 25 and 250 microseconds). The pulse frequency, which is the number of pulses per second for LAI, is typically within the range of 10–20 Hz, meaning that per second 10–20 of these very short pulses are emitted.

As erbium lasers have extremely high absorption in aqueous liquids (Wieliczka et al., 1989), the penetration depth of these wavelengths is only a few micrometres, with all the laser energy being absorbed within this limited water volume. This effect is wavelength specific: most other lasers have much lower absorption in water. The extreme absorption causes superheating of this water volume, followed by explosive boiling and formation of vapour. The vapour pressure makes the bubble expand (Figure 1b–e). During expansion, the internal pressure and the vapour temperature decrease to a point below the ambient pressure. The bubble then undergoes a forceful collapse (Figure 1f–i) (Blanken & Verdaasdonk, 2007; Matsumoto et al., 2011). It is important to realize that these events occur extremely fast. Typically the life time of a vapour bubble during LAI is in the range of a few hundred microseconds (1/1 000 000 of a second, not to be confused with milliseconds). The bubble implosion is especially fast and powerful, with speeds calculated to be several metres/second (5–21, depending on pulse energy) (Blanken et al., 2009). Hence, the liquid surrounding the bubble moves at equivalent speeds.

These laser-induced vapour bubbles are a very pure form of *cavitation*, which is the formation and violent implosion of cavities inside liquids due to some form of tension or energy deposit within the liquid (Brennen, 1995). Cavitation is a very powerful physical phenomenon. Cavitation for example poses engineering challenges with regard to pumps or ship propellers, as it can inflict serious damage to their metallic components,

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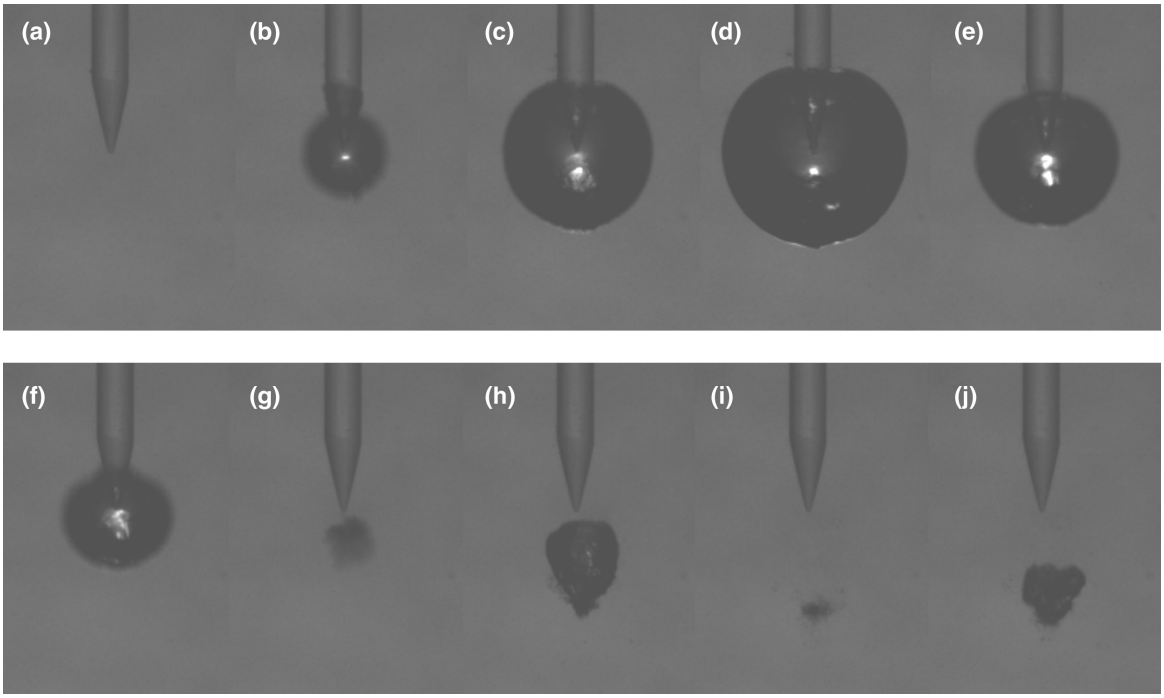

FIGURE 1 The conical tip of an Er:YAG laser in water. Growth (a–d) and implosion (e–g) of a vapour bubble at the fibre tip (this process lasts about 240 microseconds) followed by appearance of secondary cavitation bubbles (h–j).

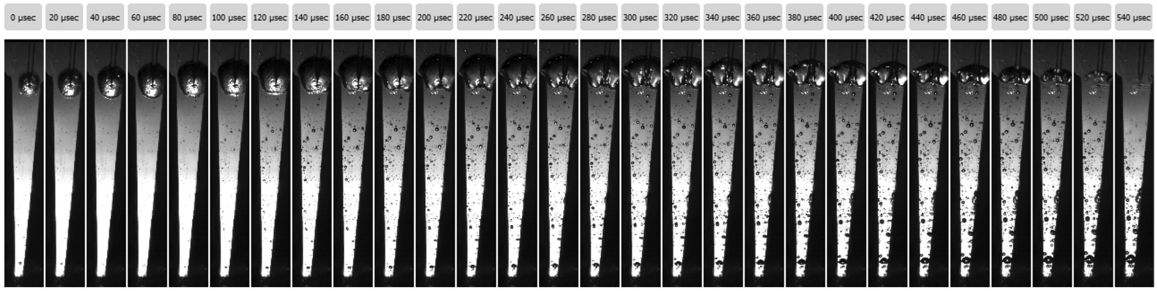

FIGURE 2 High-speed image sequence of an Er:YAG laser pulse in a root canal model. Each frame shows a root canal model which is 16-mm long and has an apical diameter of 0.35 mm. A conical 400 µm laser tip is positioned within the pulp chamber, and the canal is filled with water. Subsequent frames depict the cavitation and other events taking place within the canal after 1 laser pulse. The time in microseconds is indicated above each frame (reproduced with permission from Swimberghe et al., 2022).

or cause significant noise and severe vibrations (Zhu et al., 2021). Cavitation effects are also used in various medical laser applications such as lithotripsy, the fragmentation of bladder stones using holmium YAG lasers for example (Chen et al., 2017).

When there is sufficient space and liquid around the fibre tip, and bubble implosion can take place unimpeded, the violent collapse results in creation of a shock wave (a powerful wave of high pressure) (Gregorčič et al., 2012). In addition, secondary cavitation takes place after bubble collapse (Matsumoto et al., 2011). These are smaller bubbles that appear and disappear in an oscillating kind of rebound pattern, long (several hundred microseconds) after termination of laser pulse (Figure 1j–l).

So when short erbium laser pulses are emitted within a large liquid reservoir, extremely fast vapour bubble expansion and implosion take place, with the following consequences:

- high-speed fluid motion around the expanding and imploding bubble
- shock waves on collapse
- secondary cavitation

LAI: FROM CAVITATION TO CLEANING THE ROOT CANAL SPACE

When these lasers are operating in smaller (root canal-like) reservoirs, the situation is different. The vapour bubbles undergo friction along the cavity or canal walls, and

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there is limited available space for fluid displacement. Therefore, vapour bubbles are smaller, bubble oscillation times (the time from bubble origin to complete collapse) are longer (about 2–2.5 times) (Matsumoto et al., 2011), cavitation dynamics is slower and as a consequence, no shock waves are taking place (Lukac, Gregorcic, & Jezeršek, 2016; Lukac, Zadravec, et al., 2016).

However, other mechanisms come into play (Swimberghe et al., 2022) when working in root canal-like reservoirs. This is visualized in Figure 2, where the tip of an erbium laser is positioned within the pulp chamber of a transparent root canal model. As in a large reservoir, a vapour bubble is forming around the fibre tip. The hard cavity walls force the primary bubble to adapt to the cavity shape and the vapour bubble expands mainly coronally. Within the underlying canal, a slight apical movement (about 100 μm in distance) of the liquid is observed during this phase. The implosion of the primary bubble is not centripetal as in an infinite reservoir, but occurs in a more coronal direction. Interestingly, during this implosion, a cloud of smaller bubbles is being formed within the entire underlying root canal (Gregorčič et al., 2016). It seems that the coronally directed implosion of the primary bubble produces a negative pressure in the canal, causing existing gas inclusions to grow or generating new bubbles, thus creating cavitation throughout the canal. The result of these growing bubbles cloud is that the canal content is displaced coronally.

Upon implosion of the primary bubble, these intracanal cavitation bubbles implode as well. The net result of all this is a small yet very fast up and down movement of the canal content with each laser pulse. This happens 15–20 times per second (determined by the chosen pulse frequency), resulting in significant intracanal turbulence and liquid movement. It has been shown that this leads to cleaning of the canals walls, as both artificial and genuine microbial biofilms have been shown to be dislodged and then gradually transported coronally (Swimberghe et al., 2022). Interestingly, the cleaning pattern starts coronally and progresses apically (Gregorčič et al., 2016). Su et al. (2020), studying the fluid flow in a root canal model during LAI using particle image velocimetry, observed a back-and- forth liquid movement along the canal, confirming the above observations. They stated that LAI activates a so-called 'breath mode' during irrigation, referring to the repetitive in- and outward movement of the canal content, analogous to breathing air in and out. Koch and co-workers, also using particle image velocimetry, found that intracanal fluid speeds induced by LAI (0.3–0.5 m/s) were about 10 times higher than those obtained by an ultrasonically oscillating file. PIPS resulted in significant fluid movement up to 19 mm away from the fibre tip, while movement of liquid was mostly lacking past 1 mm below the ultrasonic tip (Koch et al., 2016). Cai et al. (2022) used high-speed imaging to determine fluid velocity in the apical region induced by LAI, and found fluid speeds ranging from 1 to 4 m/s. These studies demonstrate that erbium laser-induced cavitation bubbles in the pulp chamber or at the canal entrance result in significant irrigant movement throughout the underlying canal space. The fact that irrigant movement is generated in root canal regions which are considerably distant from the tip represents cleaning 'from a distance'. In that sense, LAI is almost unique, as most other irrigant activation methods (such as sonically or ultrasonically oscillating tips) require intracanal placement of the instrument. Only multisonic irrigation, a system generating a broad spectrum of sound waves within the irrigant in order to remove soft tissue and bacteria, has a similar remote mode of action. The independence of intracanal positioning is thought to make the procedure easier and less operator dependent.

HISTORICAL/COMMERCIAL TRENDS IN LAI

The first papers reporting on LAI were characterized by the following characteristics: the laser tips were positioned within the root canal itself (between 1 and 5 mm from working length), relatively high pulse energies (varying between 60 and 80 mJ) were used and fibres with a flat end were mostly chosen (de Groot et al., 2009; De Moor et al., 2009; De Moor et al., 2010; Deleu et al., 2015). Because it was observed that the intracanal vapour bubbles extended several millimetres beyond the fibre tip (Blanken et al., 2009; Blanken & Verdaasdonk, 2007), this approach raised concerns regarding potential extrusion of irrigating solution. Therefore, a shift towards a more coronal position took place, with at first fibre tip relocation in the coronal one third (Seet et al., 2012, Deleu et al., 2015) or within the root canal entrance (Licata et al., 2015). With the advent of the PIPS, an even more coronal laser tip position for Er:YAG lasers was proposed. The PIPS or photon-induced photoacoustic streaming (Divito et al., 2012) approach introduced a radical change in LAI. In the PIPS technique, a conically shaped laser tip is positioned within the pulp chamber (Figure 3) and operated with very short (50 μs) pulses and much lower pulse energies (10–20 mJ). The conical tip, in comparison with a flat tip, has a greater cavitation efficacy. That is, for otherwise similar laser settings, conical tips produce greater cavitation bubbles with more mechanical energy compared to flat fibre tips (Gregorcic et al., 2012). The same holds for the pulse duration: for otherwise similar laser settings, shorter pulses result in greater mechanical energy of the vapour bubble compared to longer pulses (Gregorcic et al., 2012).

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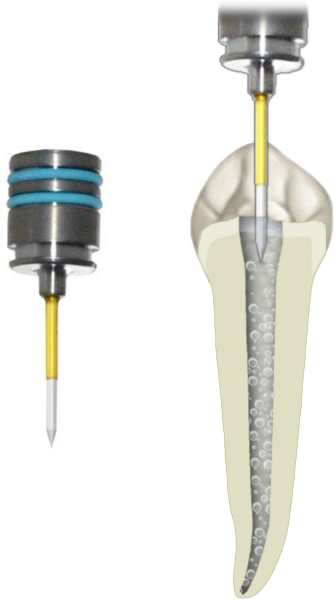

FIGURE 3 A PIPS tip (left) and its positioning in the pulp chamber during LAI (right).

Although the PIPS tip is said to be specifically designed, the 3 mm stripped part of the tip does not seem to affect its optical properties, as it is the protective sheet that has been removed, while the cladding is still intact. Hence, the tip behaves no different than similar conical fibre tips without the stripped part.

So the historical trend in LAI is from an intracanal fibre position, using flat fibre tips and relatively high pulse energies, to a more coronal position within the pulp chamber, using conical fibre tips and lower pulse energies.

Biolase, manufacturer of Er;Cr:YSGG lasers on the other hand, has a different approach and promotes intracanal laser operation. The recommended protocol for the Waterlase Er;Cr:YSGG laser is to place the fibre tip 1–5 mm short of working length. Pulses (25 mJ) are delivered while moving the fibre in coronal direction, 3–4 times per canal. This is first done in the presence of water, then using the irrigant of choice (3 times per canal and per irrigant) (https://www.biolase.com/rapidendo/).

Clearly, these represent two completely different LAI approaches. Guidelines or evidence or consensus regarding the optimal LAI parameters (such as activation time, position of the fibre tip, pulse length, pulse energy) are lacking. At the moment, most clinicians follow the manufacturer's recommendations of the laser device.

PARAMETERS AFFECTING LASER-INDUCED CAVITATION

When it comes to operating a pulsed laser in the context of LAI, there are quite a lot of parameters which can be chosen or adjusted. They include laser wavelength, pulse energy, pulse length, pulse repetition rate, irradiation time, fibre tip diameter, fibre tip geometry and fibre tip position. Many of them have a significant effect on the laser-induced cavitation, and hence on the LAI efficacy. Most important parameters will be discussed next.

Laser wavelength

The mechanism behind the cleaning action of LAI is the formation and subsequent implosion of vapour bubbles in the irrigant. For this cavitation to take place, a high-energy concentration is required to heat and vaporize a small volume of liquid in front of the laser tip. One of the major parameters determining the concentration of energy is the degree of absorption of the laser wavelength in the irrigant. Since most root canal irrigants are aqueous solutions, it is relevant to look at the degree of absorption of different wavelengths in water. The absorption characteristics of pure water in the visible and infrared spectrum are known (Hale & Querry, 1973; Wieliczka et al., 1989) and have also been tested for endodontic irrigants (Meire et al., 2014). The absorption spectrum of water (Figure 4) depicts for every wavelength the degree of absorption (given by the unit 'alpha', expressed in cm^{-1}) within water. The spectrum shows that the Er:YAG wavelength (2940 nm) coincides with the absorption peak of water around 3000 nm. The wavelength of the Er;Cr:YSGG laser (2780 nm) lies very close to this peak. Consequently, both lasers have extremely high alpha values: 13 000 and 7000 cm^{-1} respectively.

When considering the spectral region of the infrared diode lasers, ranging from 810 to 1064 nm, alpha values of less than 0.2 cm^{-1} are the case, demonstrating a tremendous difference in absorption between diode and erbium laser wavelengths.

Apart from the degree of absorption in the liquid, other parameters play with regard to the energy concentration. Concentration of energy can also be obtained temporally (focusing the energy within a very short time), as well as spatially (by concentrating the energy on a very small area). For example, Lauterborn and Ohl (1997) used an Nd:YAG laser (1064 nm) which has very low absorption in water (alpha: 0.1 cm^{-1}). They managed to produce cavitation bubbles in water by using very short (8 ns) laser pulses and extreme focusing of the beam with lenses. When operating within the endodontic space, spatial energy concentration is dictated by the fibre tip diameter. However, the maximum power allowed by smaller tips is usually lower than that of wider tips, so spatial focusing is of little gain. The temporal

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concentration depends on the shortest pulse that the laser system can deliver, which, for dental lasers, mostly is in the microsecond range, limiting temporal focusing. So the degree of absorption in the liquid of the laser wavelength itself remains the most important factor in determining the cavitation efficiency. In this respect, a minimal absorption coefficient value of 10 cm^{-1} has been put forward with regard to the activation of irrigants (Meire et al., 2014). This means that diode lasers are not suited for cavitation-based LAI, though some authors (Hmud et al., 2010) and companies state otherwise. Therefore, nowadays among lasers available for dental application, erbium lasers (Er:YAG and Er;Cr:YSGG) and to a lesser extent the Nd:YAP laser (1340 nm) are suited for this purpose. When experimentally determining the vapour bubble size and resulting fluid movement of these three wavelengths, the Er:YAG was found to be the most effective in cavitation bubble formation and in generating fluid motions within narrow lateral canals (Lukac, Zadravec, et al., 2016).

Pulse energy

Figure 5 depicts the influence of pulse energy on vapour bubble size, based on high-speed imaging measurement of cavitation in water, demonstrating the directly proportional relationship between pulse energy and vapour bubble size. Similar findings have been reported by Lukač et al. (2016). The fact that higher input of laser energy results in greater cavitation bubbles is quite logical. With regard to clinical application, the pulse energy has to be balanced against practical considerations as the amount of available liquid in the pulp chamber and safety issues with regard to pressure and temperature.

Pulse length

Just like pulse energy, the pulse duration has a marked effect on the erbium laser-induced cavitation. The relationship is however inversely proportional: when keeping the pulse energy constant, shorter pulses result in

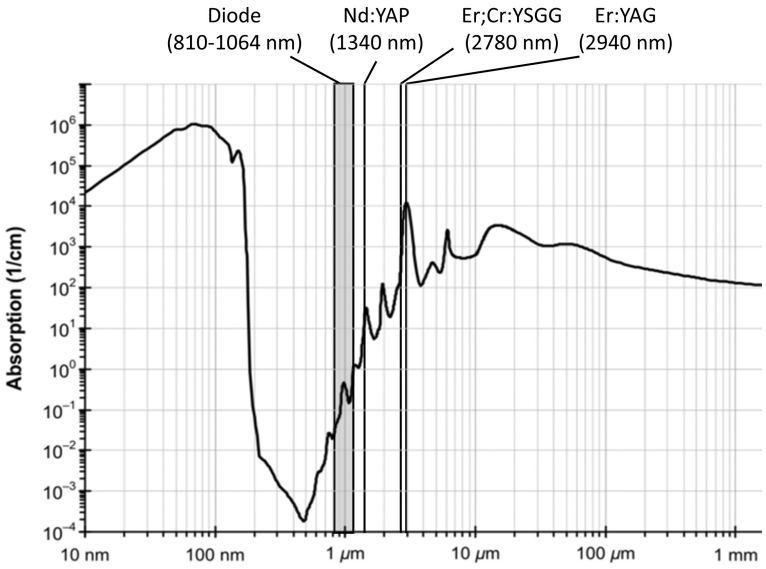

FIGURE 4 The absorption spectrum of water with selected dental laser wavelengths.

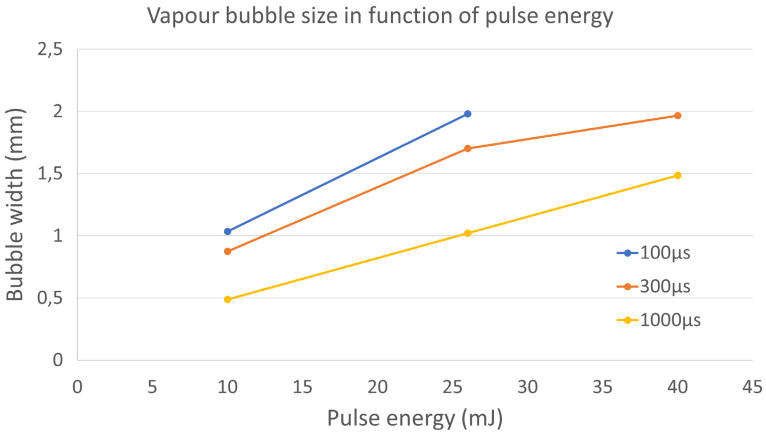

FIGURE 5 The influence of pulse energy on vapour bubble size for different pulse lengths.

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larger vapour bubbles, as shown in Figure 6. Measuring the vapour bubble dynamics for various parameters using shadowgraphy, Gregorčič et al. (2012) reported similar findings, namely that shorter pulses produce bubbles of greater volume. So in order to obtain highest cavitation efficiency, the smallest pulse duration seems the option of choice. This parameter however differs between laser devices, with the smallest available pulse durations to date varying between 25 (Lightwalker, Fotona) and 200 (AdvErL EVO, Morita) microseconds.

Fibre tip geometry

Basically, two fibre tip geometries exist for LAI application, that is, a flat and a conical one. The consequence of a different geometry is a completely different emission pattern of the two fibres. In a conical fibre tip, the light undergoes several reflections in the conical end before it exits (Figure 7). The result is twofold: the light is being concentrated in the very distal part of the conical tip so the energy is concentrated in a small spot, and light exits the fibre tip at different lateral angles. In flat fibre tips, the light exits the fibre tip in a more or less straightforward direction, with minimal lateral emission (a divergence of 10–15 degrees).

So compared to flat fibre tips, the energy density at conical tips is much higher, and their emission pattern is more lateral (George & Walsh, 2009). As a result, vapour bubble formation around flat and conical fibre tips is different as well: flat tips produce ellipsoidal bubbles, while conical fibre tips produce spherical bubbles (Gregorčič et al., 2012) (Figure 8).

Consequences for clinical application

From the above, it appears that for optimal cavitation efficiency, erbium lasers should be used, equipped with conical fibre tips, emitting the shortest possible pulses with high pulse energy.

In a laboratory study exploring the influence of various lasing parameters on the cleaning efficacy of LAI, Meire et al. (2016) evaluated the influence of pulse energy, pulse frequency, pulse length, irradiation time and fibre tip shape and position and diameter on the removal of dentinal debris from a lateral groove in a root canal model. It was found that shorter pulses, higher pulse energy, higher frequency, longer irradiation time and position of the fibre tip close to the groove were correlated with better debris removal. Hoshihara et al., assessing removal of $Ca(OH)_2$ paste from a lateral canal in a root canal model, came to similar conclusions. They found improved $Ca(OH)_2$ removal with increasing pulse energy and fibre tip insertion closer to the lateral canal. No significant influence of pulse frequency was found. In their model, 600-μm wide tips performed better than 200-μm wide tips (Hoshihara et al., 2021). Jezersek et al. (2021) on the other hand found that the removal rate of simulated debris by flat fibre tips outperformed that of the radial fibre tip. The experimental model however was very basic with a 0.8-mm diameter cylindrically shaped root canal model which was filled entirely with tooth paste.

DOUBLE-PULSE LAI: SWEEPS

As mentioned previously, shock wave emission following implosion of the primary cavitation bubble is observed in a free water environment, but not in confined reservoirs such as pulp canal systems (Lukac & Jezersek, 2018). In order to enhance the emission of shock waves from collapsing bubbles in endodontic spaces, a new laser modality called SWEEPS™ (shock wave enhanced emission photoacoustic streaming) was introduced (Lukac et al., 2017). The technique is based on the delivery of pairs of Er:YAG laser pulses, with the second pulse timed in such a manner to intensify the collapse of the bubble generated by the

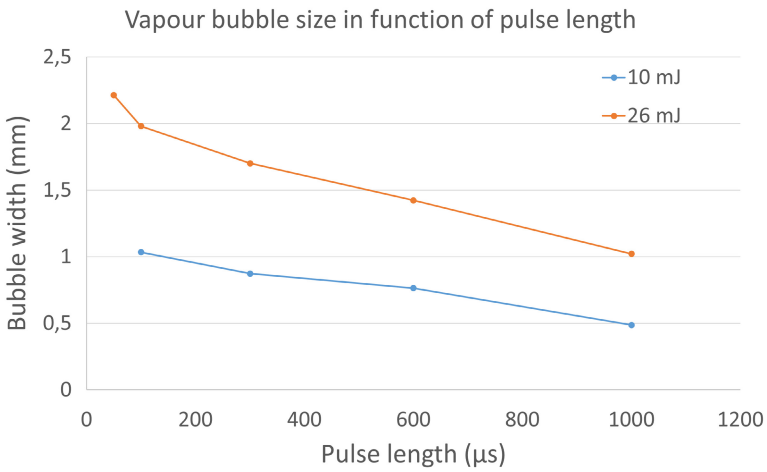

FIGURE 6 The influence of pulse length on vapour bubble size for different pulse energies.

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first pulse, thereby producing shock waves and enhanced irrigant streaming also in confined spaces (Figure 9).

The timing of the second bubble of the pair is crucial, as the SWEEPS effect does not take place when the second bubble is delivered too early or too late. It has been experimentally determined that the optimal delay between the two laser pulses is when the second bubble starts to develop near the end of the first bubble's collapse (Lukac & Jezersek, 2018). However, oscillation times of cavitation bubbles vary considerably as they depend strongly on pulse energy and on cavity diameter (being longer for smaller cavities and/or for higher laser pulse energies). In order to cope with this variability, the AutoSWEEPS modality was developed (Lukac et al., 2017), in which the temporal separation between the pulse pairs is continuously swept back and forth in order to ensure that during each sweeping cycle there is always at least a 50-μs wide temporal range where the optimal separation between the pulse pair is achieved, as required for shock wave generation. The sweeping modality also ensures that the optimal conditions are approximately reached along the depth of the access cavity by matching the changing diameter conditions during the AutoSWEEPS cycle. It was also observed that when the bubble pair separation time was shorter than the optimal separation time, the second bubble effectively 'propels' the initial bubble deeper into the canal, thus effectively improving the irrigation effect deeper within the canal (Lukac et al., 2017).

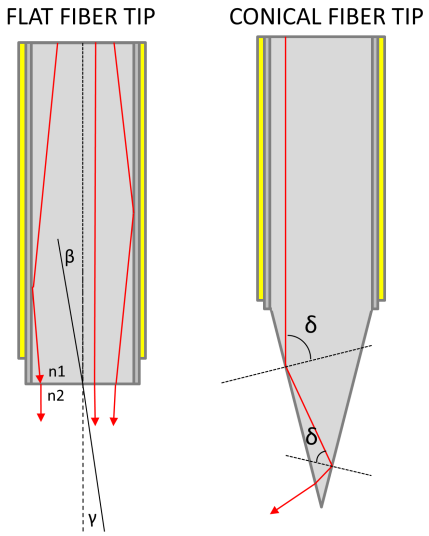

FIGURE 7 Difference in emission pattern between flat and conical fibre tips. In a flat fibre tip, the exit angle γ is determined by the critical angle for internal reflection β (which is 9 degrees for silica fibres), and the refractive indices of the core (n1) and the surrounding (n2) (Snell's law: $n1 \sin \beta = n2 \sin \gamma$). In a conical fibre tip, the light undergoes several reflections in the conical end. As the angle of incidence δ decreases, the light will not only be reflected, but also refracted, thereby causing the light to exit the tip. As a result, the light gathers at the bottom part of the conical FT and exits the FT at different angles. Light grey represents the fibre tip core, darker grey the cladding, yellow the jacket.

Two clinical studies addressed the single- versus double-pulse LAI. Erkan et al. (2022), in a randomized controlled trial investigating postoperative pain after root canal treatment of mandibular premolar teeth with symptomatic irreversible pulpitis and different activation methods, found significantly lower postoperative pain scores after LAI systems compared to the other activation systems, but without a difference between PIPS and (auto-) SWEEPS. Mittal et al. (2023), in a similar setup, found lower postoperative pain scores in LAI systems as compared to the other activation systems, without significant differences between PIPS and SWEEPS.

The majority of head-to-head comparisons of single- versus double-pulse LAI however are laboratory studies. Galler et al. (2019) compared penetration depths of endodontic irrigants into the dentinal tubules of extracted teeth with several activation methods, including single- and double-pulse LAI. They found that irrigant penetration with PIPS

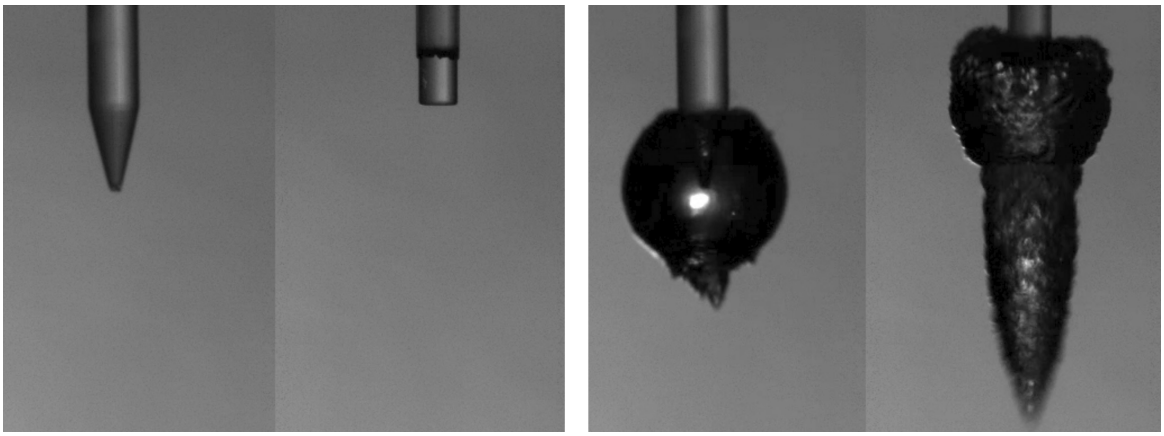

FIGURE 8 Conical and flat fibre tip before (left) and during (right) emission of an Er:YAG laser pulse, resulting in a more or less spherical bubble around the conical tip, and an ellipsoid bubble at the flat tip.

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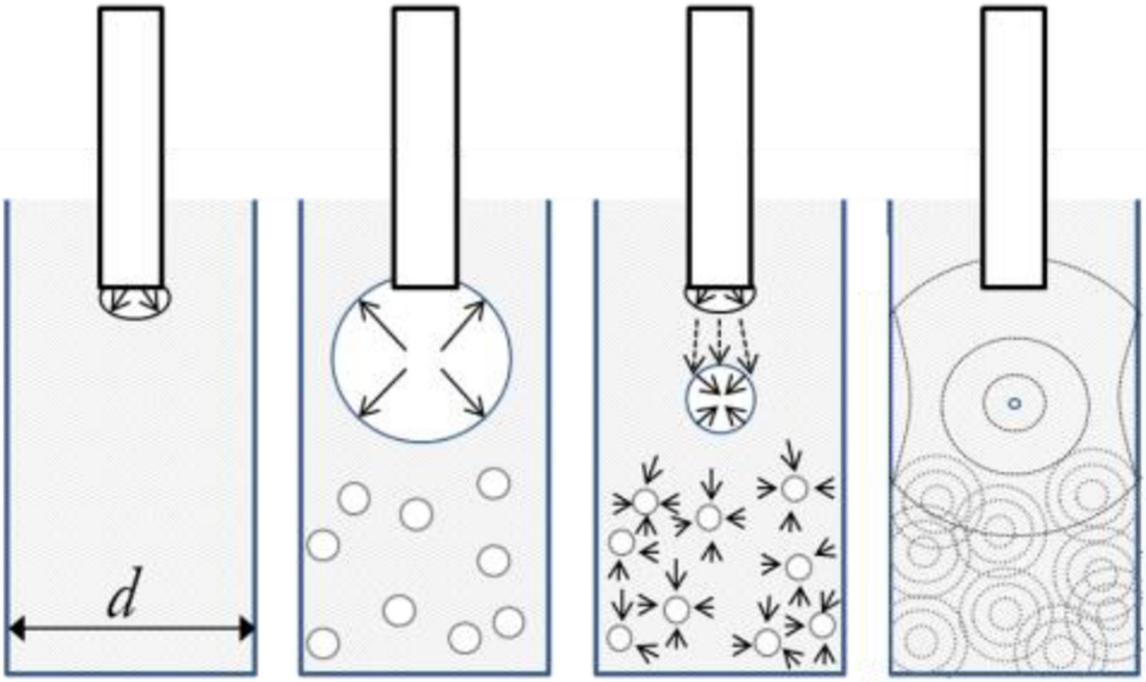

FIGURE 9 Principle of double-pulse LAI, in which a second laser pulse is generated (c) with a certain delay, accelerating the implosion of the primary bubble and of the smaller bubbles that have formed secondary to it. Reproduced with permission from Lukač et al. 2017 (Copyright 20 127 LA&HA, Laser and Health Academy).

was significantly deeper than that with SWEEPS. Kosarieh et al. (2021) on the other hand found no differences between the two modalities with regard to dye penetration in the dentine. Mancini et al. (2021) found no differences in smear layer removal between PIPS and SWEEPS activation. Yang et al. (2020) found that removal of accumulated hard tissue debris from canal systems of lower molars with SWEEPS was greater than that with PIPS (and UAI), especially in isthmus-containing mesial roots.

Petricevic et al. used micro-CT analysis to assess removal of root canal filling remnants after conventional re-treatment in severely curved molar root canals. They found significantly higher removal of obturation remnants with single-pulse LAI (600 μm conical tip in the pulp chamber, 20 mJ, 15 Hz, 50 μs) compared to SWEEPS with a flat tip. When SWEEPS was used with a conical tip, the difference was no longer significant (Petricevic et al., 2022). Kirmizi et al. (2021), evaluating calcium hydroxide removal from simulated internal resorption cavities using micro CT, found no differences between PIPS and SWEEPS. Jezersek et al. (2021) evaluated debris removal from an artificial root canal and reported better performance of double-pulse in comparison with single-pulse LAI. As mentioned previously, the experimental model was rather weak. In summary, it seems that double-pulse LAI is an effective LAI modality, but evidence for improved efficacy over single-pulse LAI seems to be lacking at the moment.

LASER-ACTIVATED IRRIGATION: CLINICAL EVIDENCE

While there is a substantial body of *in vitro* research, there are very few clinical studies investigating LAI. Regarding the effect of LAI on healing of apical periodontitis, there is only one randomized clinical trial comparing conventional chemomechanical preparation to Er;Cr:YSGG laser-assisted endodontic treatment (Martins et al., 2014). In this study, single-rooted and premolar teeth were treated in two visits, in an undergraduate student setting. All teeth received conventional chemomechanical preparation. In the control group, NaOCl was used as irrigant and there was an interim $Ca(OH)_2$ dressing. In the laser group, the irrigant was physiological saline, and after canal preparation intracanal Er;Cr:YSGG laser irradiation (both in a dry canal + filled with distilled water) was performed. No interim dressing was used. No significant difference in apical healing between both groups was found after 1 year (Martins et al., 2014).

A few clinical trials have looked at postoperative pain following LAI. This has been a concern as the very powerful irrigant dynamics have been thought to constitute an increased risk of irrigant extrusion and higher levels of postoperative pain. In addition, a number of laboratory studies have demonstrated increased apical pressure (Peeters & De Moor, 2015) and increased irrigant

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extrusion in laboratory models (Azim et al., 2018; George & Walsh, 2008) with LAI. A number of randomized clinical trials have investigated the incidence and degree of postoperative pain after root canal treatment with adjunctive LAI versus other activation techniques (Dagher et al., 2019; Erkan et al., 2022; Liapis et al., 2021; Mandras et al., 2020; Mittal et al., 2023). None of these studies found a higher level of postoperative pain after LAI versus no or other activation methods, nor any untoward events using LAI. Mandras and colleagues in their clinical trial also measured intracanal microbial load in both groups (needle irrigation vs. LAI). They found no significant difference in bacterial reduction among the groups. However, when looking at mean bacterial reductions in the needle irrigation group (90–98%), these values are not conform figures found in typical studies (e.g. Vianna et al., 2006: mean reductions of 99.9%), which questions the microbiological method in that study.

So while it has been clinically demonstrated that LAI application does not result in higher levels of postoperative pain, there is little clinical evidence demonstrating added value with regard to root canal cleaning and disinfection, or improved healing rates. This has mainly to do with a lack of clinical trials addressing these outcomes. In this respect, 'absence of evidence' does not necessarily mean 'evidence of absence'. To put this into perspective, a recent systematic review looking at all types of adjunct therapy (any type of extra canal cleaning/disinfection step, including sonic and ultrasonic activation, photo-activated disinfection, direct laser irradiation, etc.) concluded that there was insufficient evidence to recommend any adjunctive therapy for the treatment of apical periodontitis (Meire et al., 2022). This was not due to lack of a demonstrable effect, but mainly due to the lack of qualitative trials assessing the efficacy of these adjunct therapies. So as is the case with many topics in endodontics, high-quality evidence regarding the clinical effectiveness of various therapies or treatment aspects is lacking. The same is true for LAI.

LAI: LABORATORY STUDIES ON ROOT CANAL DISINFECTION

A number of review articles have addressed disinfection and biofilm removal with LAI (Badami et al., 2023; Do & Gaudin, 2020; Josic et al., 2022). Do and Gaudin (2020) reviewing the literature on the effectiveness of Er:YAG LAI versus other endodontic irrigation methods and techniques in extracted teeth, found that with regard to canal disinfection, Er:YAG LAI outperformed other methods in 11 out of the 18 studies, with no significant differences between methods in the other 7. They concluded that Er:YAG LAI is promising in canal disinfection as well as debris and smear layer removal (Do & Gaudin, 2020). They also mentioned a great variety in study designs, reducing the significance of the results. In a narrative review on *in vitro* endodontic biofilm models and efficacy of sonic-, ultrasonic- and laser-assisted biofilm removal techniques, Josic et al. (2022) concluded that LAI as well as ultrasonic and sonic irrigant activation provide a more efficient biofilm removal compared to conventional syringe/needle irrigation, and that due to heterogeneity in methodologies, it is difficult to compare adjunct therapies and recommend the most efficient method in biofilm removal. Badami et al. (2023), conducting a systematic review on the efficacy of LAI versus ultrasonic activation in extracted permanent teeth, found that LAI showed superior antimicrobial efficacy than UAI in five out of eight selected studies, while in the other three no significant difference between the two techniques was found. They concluded that LAI not only has superior efficacy over UAI in the elimination of microorganisms (Badami et al., 2023), but also mention the problematic variability in methodology among studies. This variability includes the methods for biofilm cultivation, the dimensions and taper of the canal, irrigant type, irrigant concentration, irrigation time, needle design, laser settings, tip design and tip insertion depths.

In order to overcome the lack of standardized protocols and discrepancy in the methodology and select the best possible evidence, we applied the following selection criteria, taking into account considerations expressed in Swimberghe et al. (2019), Swimberghe et al. (2021) and Meire and van der Waal (2022) regarding biofilm model systems for root canal disinfection and research methods to study intracanal disinfection:

- The substrate for biofilm development or infection simulation should be dentine, not an artificial substrate
- The sample size should be no less than eight per group, including the untreated control group
- The biofilm or root canal infection should have no less than 7 days of incubation
- LAI parameters should be reported in sufficient detail (including pulse energy, pulse length, tip design and dimensions), and irrigant activation time should be no less than 30 seconds
- Harvesting/sampling after treatment: an active sampling method should be used (e.g. scraping of canal walls or flushing under pressure), the mere collection of canal fluid using paper points is insufficient
- The quantification method should be accurately described, including results of the untreated control group
- The antimicrobial effect should be assessed through comparison between (LAI-)treated groups and the untreated control group. Within-tooth comparison

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between a microbial sample taken before and after treatment is deemed inadequate
- When the plate count method is used, CFU/sample or CFU/biofilm should be reported instead of CFU/mL [Also, it is appropriate to report the original microbial counts (possibly log transformed) from all groups instead of only giving the logarithmic or the percentage reductions with regard to the control.]
- For LAI, there should be a four-wall cavity/reservoir that is kept full with irrigant throughout the LAI activation

Critical appraisal of the available literature on the antimicrobial efficacy of LAI resulted in selection of six papers, which are summarized in Table 1.

In the study by Peters et al., prepared extracted teeth were worn intraorally by volunteers, cultivating intracanal microbial biofilms. After 2 more weeks of incubation in the laboratory, the teeth were subjected to 6% NaOCl with syringe irrigation, ultrasonic activation or a LAI using the PIPS approach. After sampling, microbial quantification took place using the plate count method, the incidence of positive samples was determined and histobacteriological analysis of root sections 1 mm and 4 mm from the apex was done. LAI yielded more negative bacterial samples and left less apical bacteria/biofilm than ultrasonic activation (Peters et al., 2011). Another comparison between LAI and UAI showed superior dentinal tubule disinfection for PIPS application (Neelakantan et al. 2015). In this study a 4-week *Enterococcus faecalis*biofilm was grown in prepared single-rooted mandibular premolars. Ultrasonic activation, diode laser (940 nm) irradiation and PIPS were evaluated in combination with (i) 3% NaOCl with 9% etidronic acid, (ii) 3% NaOCl and 18% EDTA and (iii) 3% NaOCl, 17% EDTA and a final flush of 3% NaOCl. Confocal laser scanning microscopic examination was used to assess bacterial viability *in situ*. Furthermore, dentine debris from the root samples was harvested at depths of 200 and 400 microns in order to determine colony-forming unit counts. The use of NaOCl after or in combination with a chelator caused the greatest reduction of *E. faecalis*. Diode laser and PIPS approach were superior to UAI in dentinal tubule disinfection. The laser settings were identical to the study of Peters et al. (2011). So, both studies have in common that better disinfection can be achieved with PIPS than with UAI. The laser settings are different from the present-day advocated ones: 50 mJ instead of 20 mJ. In the study of Neelakantan et al., saline served as the control.

Cheng et al. evaluated the bactericidal effect of PIPS LAI in NaOCl on *E. faecalis* biofilms in two studies (Cheng, Tian, et al., 2017; Cheng, Xiang, et al., 2017). In Cheng, Xiang, et al. (2017), prepared straight roots were contaminated for 4 weeks with a clinical *E. faecalis* isolate with the strongest biofilm formation abilities, taken from root-filled teeth with peri-apical lesions. Syringe irrigation, ultrasonic activation or LAI using the PIPS approach with normal saline or 5.25% NaOCl was compared. Dentin shavings were collected for microbial quantification using the plate count method. No significant difference in bacterial reduction was observed between UAI and PIPS with NaOCl. In the second study of Cheng, Tian, et al. (2017), prepared permanent teeth with straight root canals were contaminated for 4 weeks with an *E. faecalis* strain. Canals prepared to different apical diameters though with the same taper were exposed to NaOCl PIPS LAI (including different activation times and pulse energies) or syringe irrigation with 5.25% NaOCl. Bacterial reductions were evaluated using the plate count method. The differences in apical preparation diameters make the study more interesting, as well as the investigation of the increase in average power. In each situation, PIPS LAI outperformed conventional needle syringe irrigation for identical apical canal diameters. In fact, to achieve similar disinfection efficacy, the PIPS NaOCl group needed a smaller diameter than the NaOCl group (the disinfection efficiency of the #15 PIPS LAI was similar to that of #40 NaOCl SI). Longer irradiation times and higher output power increased the disinfection efficacy of PIPS LAI. In both studies of the PIPS, tip was positioned 1 mm below the orifice of the canal and activation cycles of 15 sec were relied upon. It has to be mentioned that this is not the recommended present-day approach for PIPS LAI.

Only two studies with Er,Cr:YSGG LAI were selected (Betancourt et al., 2019; Betancourt et al., 2020). In those studies, prepared single-rooted teeth were inoculated with *E. faecalis* for 10 days. In the first study, the antibacterial effectiveness of Er,Cr:YSGG activation of 0.5% NaOCl and saline was compared to syringe irrigation with 0.5%, 2.5%, 5% NaOCl and saline (Betancourt et al., 2019). Microbial quantification took place using the plate count method and was completed with scanning electron microscope (SEM) and confocal laser scanning microscopy. In the second study, the antibacterial effectiveness of Er,Cr:YSGG LAI and UAI were compared using both 0.5% NaOCl and saline irrigant (Betancourt et al., 2020). The number of bacterial survivors was determined by plate counting. The data were completed with atomic force microscopy imaging and SEM. In both studies intracoronal activation of the irrigant took place, while the recommended Er;Cr:YSGG protocol according to the manufacturer is an up- and downwards movement of the fibre tip in the root canal starting close to the working length and ending in the coronal third. In order to ensure adequate irrigant agitation in this way, higher energy is required (the classic intracanal approach relies on 25 mJ pulses). In the first study, the investigators demonstrated that the use of 100 mJ pulses (higher than the classic approach) and tip

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TABLE 1 Experimental characteristics of selected disinfection studies.

Author/year	Activation methods	Infection model	Number and length of root samples	Canal preparation size and diameter	Laser settings (pulse energy, repetition rate, pulse length) and laser tip	Activation time/protocol	NaOCl	Microbiological evaluation	Main findings
Peters et al. (2011)	PIPS, UAI, SI	Biofilm grown in sterilized extracted teeth attached to oral devices (6–8 days) followed by 15 days *in vitro* incubation	4 groups of 15 roots (14 mm long)	20/0.07	50 mJ, 10 Hz, 50 μs 400 μm, 21 mm (PIPS tip)	PIPS: 30 s activation, 30 s rest time UAI: 60 s SI: 60 s	6%	Plate count method Histologic evaluation (Brown–Brenn staining)	PIPS generated more negative bacterial samples and less apical bacteria/biofilm tan UAI
Neelakantan et al. (2015)	PIPS, UAI, diode laser, SI	4w *Enterococcus faecalis* (ATCC 29212) biofilms in single-rooted extracted teeth	16 groups of 20 teeth (18 mm long)	25/0.06	50 mJ, 10 Hz, 50 μs 400 μm, 21 mm (PIPS tip)	30, 60 and 90 s	3%, 6%	Plate count method Confocal laser scanning microscopy	Diode laser and PIPS were superior to UAI and no activation in dentinal tubule disinfection
Cheng, Tian, et al. (2017)	PIPS, UAI, SI	4w *E. faecalis* (clinical isolate) biofilm in single-rooted extracted teeth	6 groups of 15 roots (12 mm long)	40/0.04	20 mJ, 25 Hz, 50 μs 300 μm, n/a (PIPS tip)	PIPS: 30 s UAI: 60 s	5.25%	Plate count method SEM	Higher bacterial reduction with PIPS and UAI than with SI No difference between PIPS and UAI
Cheng, Xiang, et al. (2017)	PIPS, SI	4w *E. faecalis* (ATCC 29212) biofilms in single-rooted extracted teeth	16 groups of 20 roots (12 mm long)	15/0.04 20/0.04 25/0/04 30/0.04 40/0.04	20 mJ, 15/25/50 Hz, 50 μsec 300 μm, n/a (PIPS tip)	20 sec, 40 sec and 60 sec	5.25%	Plate count method SEM	The bacterial reduction increased as the MAFs were enlarged from #15 to #40 for both PIPS and CI PIPS significantly enhanced the disinfection efficacy of NaOCl at each apical preparation diameter The disinfection efficiency of the 15#/Er:YAG + NaOCl group was similar to that of the 40#/NaOCl SIgroup
Betancourt et al. (2019)	Er,Cr:YSGG LAI, SI	10d *E. faecalis* (ATCC 29212) biofilms in single-rooted extracted teeth	7 groups of 13 roots (14 mm long)	45/0.02	100 mJ, 10 Hz, 140 μs 200 μm, 21 mm (RTF2 tip)	Er,Cr:YSGG: 30 s activation 30 s rest time 30 s activation SI: 60 s	0.5% 2.5% 5%	Plate count method SEM Confocal laser scanning microscopy	Bacterial reduction with 0.5% NaOCl + Er, Cr:YSGG was significantly higher than with 0.5% NaOCl + SI Treatment with 0.5% NaOCl + Er, Cr:YSGG was equally effective as 2.5% NaOCl + SI

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TABLE 1 (Continued)

Author/year	Activation methods	Infection model	Number and length of root samples	Canal preparation size and diameter	Laser settings (pulse energy, repetition rate, pulse length) and laser tip	Activation time/protocol	NaOCl	Microbiological evaluation	Main findings
Betancourt et al. (2020)	Er,Cr:YSGG LAI, UAI	10 d *E. faecalis* (ATCC 29212) biofilms in single-rooted extracted teeth	6 groups of 12 roots (14 mm long)	55/0.02	55 mJ, 10 Hz, 60 μs 200 μm, 21 mm (RTF2 tip)	Er, Cr: YSGG: 30 s activation 30 s rest time 30 s activation UAI: 60 s	0.5%	Plate count method AFM imaging SEM	Er,Cr:YSGG LAI was more effective than UAI against 10-day-old intracanal *E. faecalis* biofilms

Abbreviations: PIPS, photon-induced photoacoustic streaming; UAI, ultrasonically activated irrigation; SI, syringe irrigation; SEM, scanning electron microscopy.

positioned in the pulp chamber reached the same level of disinfection than SI with 2.5% NaOCl (Betancourt et al., 2019). In their second study, activation of NaOCl with Er,Cr:YSGG laser resulted in higher bacterial reduction of 10-day-old intracanal *E. faecalis* biofilms than with ultrasonics (Betancourt et al., 2020).

As a result of the exclusion criteria of our search, studies comparing Er,Cr:YSGG LAI and PIPS LAI were not selected. The findings of Betancourt et al. (2019, 2020) as compared to Peters et al. (2011) and Neelakantan et al. (2015) might suggest a lower efficacy in biofilm interaction for Er,Cr:YSGG. It also has to be emphasized that the PIPS protocols used in the selected studies differ from the so-called 'classic PIPS protocol'. This classic PIPS protocol/final rinse protocol (Jaramillo, 2015) consists of three cycles of 30-second activation with NaOCl followed by three rounds of 30-second off cycles. A 30-second activated irrigation with water only preceded a 30-second cycle with EDTA (activation followed by resting time). Finally a 30-second cycle with PIPS and water was applied before drying of the canals and filling of the root canal system. The effectiveness of this protocol was first scientifically tested by Lloyd et al. (2014), finding greater debris removal from prepared mesial root canals of mandibular molars by micro-CT after PIPS versus after needle irrigation. Lloyd et al. later tested this protocol for the removal of calcium hydroxide from Weine type II systems. The PIPS approach showed significantly better removal of sticky $Ca(OH)_2$ paste in the apical third than UAI (UltraEndo) and needle-syringe irrigation (Lloyd et al., 2016). Modification of this protocol was suggested by Golob et al. (2017) (i) by introducing PIPS with EDTA prior to NaOCl irrigation, removing the mineralized part of the smear layer, opening dentinal tubules, hence enabling deeper penetration of NaOCl; (ii) 3×30 s activation of 5% NaOCl for the reduction of microorganisms in dentinal tubules and the denaturation of collagen, which is important for the bacterial binding, including *E. faecalis*; (iii) distilled water for the inactivation of oxygen before obturation to avoid a chemical interaction of sealer with oxygen; (iv) 30 s of activation (both EDTA and NaOCl) were followed by a resting phase of 60 s, enabling the impact of the higher reaction rate of NaOCl; (v) laser energy was reduced by 50% in order to increase the safety of the PIPS application (no difference was found in biofilm removal between higher and lower power settings).

Minimally invasive endodontics aims to minimize structural alterations after mechanical preparation of the canal system. Cleaning and disinfection at smaller tapers and smaller apical diameters of the root canal preparation might pose a number of disadvantages such as limited irrigant penetration, needle wedging, the vapour lock effect and challenges associated with the instruments used during sonic and ultrasonic irrigation (Vieira et al., 2020).

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PIPS with its three-dimensional spreading of the irrigation solution might have a positive effect on removal of bacteria when decreasing taper and diameter of the root canal. In the investigation of Cheng, Tian, et al. (2017), PIPS and 5.25% NaOCl showed a higher disinfection efficacy at each apical preparation diameter compared with SI with NaOCl alone. Under the condition of the similar disinfection efficacy, PIPS allowed a smaller apical preparation diameter compared to the SI group (the disinfection efficiency of the #15/PIPS group was similar to that of the 40#/SI group). The study of Wen et al. (2021) was not selected because lack of a negative control group showed following findings: increase of the width from #10 to #25 improves the bactericidal effect of PIPS. Increasing the taper of the root canal from 0.02 to 0.06 at size 25 did not affect bactericidal effects of PIPS. PIPS resulted in more dead bacteria in specimens with smaller tapers and root canal widths than SI.

In conclusion, after critical selection of studies assessing the antimicrobial efficacy of LAI (Table 1), the heterogeneity of the data has not resolved completely, but it can be stated that canal disinfection with LAI is superior to that of conventional irrigation, and there is a clear trend of higher antimicrobial efficacy of LAI compared to ultrasonic activation.

OTHER RELEVANT ASPECTS WITH REGARD TO LAI

Apical vapour lock

Peeters et al., using high-speed imaging, reported the removal of entrapped air from the apical canal region by Er;Cr:YSGG laser activation. It was observed that at the irrigant–air interface, irrigant droplets were propelled beyond the interface with every laser pulse, thereby filling the air column. Remaining apical air bubbles were subsequently broken into progressively smaller bubbles, until the entire canal was filled with irrigant (Peeters et al., 2015). Thus, it seems that LAI is an effective method of resolving the apical vapour lock issue. Figure 10 depicts the breaking down of large air bubbles in a root canal model under pulsed Er;Cr:YSGG laser activation.

Effect on dentin surface and dentin properties

Regarding effect on dentin hardness, Wu et al. (2021) observed no significant difference in dentinal microhardness between Er:YAG activated and nonactivated solutions. This is in agreement with Akbulut and Terlemez (2019), who found that root canal dentin microhardness was significantly affected by the irrigation solution, but not by the PIPS activation. Widbiller et al. (2023) found that LAI activation of both NaOCl and EDTA led to pronounced surface disintegration of root canal dentine, which was higher than that of sonic (Eddy) or ultrasonic activation. Hancerliogullari et al. (2021) investigated the effect of different irrigating solutions and activation techniques (LAI and PUI) on the release of different growth factors from the dentine of extracted premolar teeth. Highest growth factor release was observed in the Er:YAG activation group, albeit the difference with PUI was not significant for all growth factors.

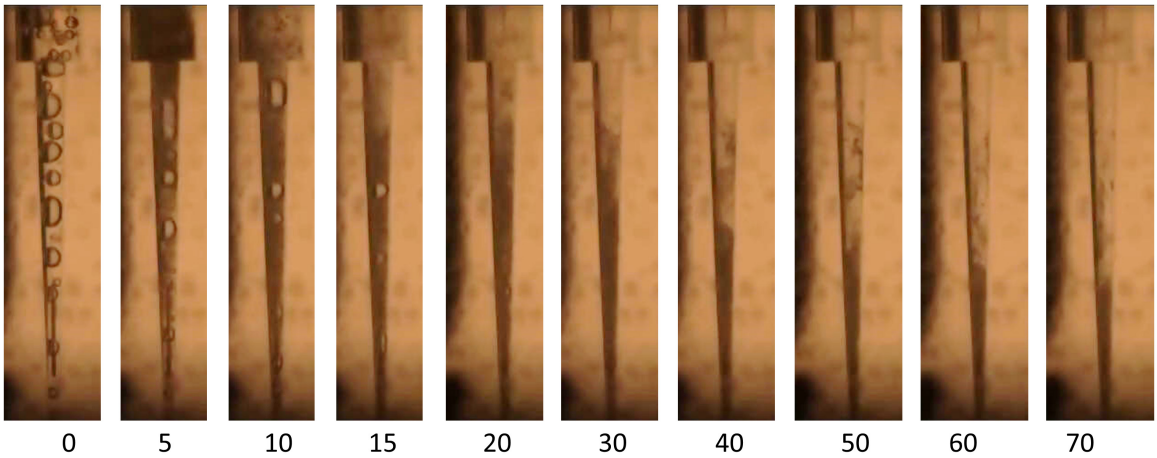

FIGURE 10 The breaking down of large air bubbles in a root canal model under pulsed Er;Cr:YSGG laser activation. The laser tip is positioned within the artificial pulp chamber. Air bubbles are progressively broken into smaller bubbles until the canal is free of bubbles. The figure below each frame depicts the number of laser pulses since the start. Images in collaboration with R. Verdaasdonk, University Medical Centre, Utrecht, The Netherlands.

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Debris removal

Studies evaluating removal of all kinds of intracanal debris or remnants (including root filling material, calcium hydroxide remnants, dentin debris, biofilm-mimicking hydrogel and pulp remnants) by LAI are so abundant that this topic warrants a separate review article.

Tissue dissolution

Liu et al. compared the dissolution of bovine muscle tissue upon exposure to irrigant activation with Er:YAG and Er,Cr:YSGG lasers, Gentlewave, PUI and CNI and found that Gentlewave and both erbium lasers achieved a significantly faster tissue dissolution rate than PUI and CNI. Tissue dissolution by Gentlewave was significantly faster than that of erbium lasers, but the amount of NaOCl in the Gentlewave test group was significantly greater as well (Liu et al., 2023). Similar findings were reported by Guneser and colleagues. They measured bovine pulp tissue dissolution by 5.25%NaOCl in test tubes after 5 minutes of activation by the EndoActivator and 2 Er:YAG laser activation protocols. Tissue dissolution by Er:YAG LAI was significantly higher compared to no activation. The Er:YAG protocol using 50 mJ pulses was significantly better than all other groups, including the 20 mJ pulse Er:YAG group (Guneser et al., 2015).

NaOCl reaction rate

In studying the reaction rate of NaOCl, Macedo et al. found that both PUI and LAI strongly increased the NaOCl reaction (the consumption of available chlorine) rate compared to no activation. LAI however sustained a higher NaOCl reactivity in the rest interval after activation than PUI (Macedo et al., 2010). The causes of such difference, however, are not yet fully understood. Sonochemical characterization of cavitation generated by ultrasound has been the subject of various papers (Dashtimoghadam et al., 2020; Macedo et al., 2014). However, similar information regarding LAI is missing. This calls for further exploration of this topic.

Temperature changes in the root

When looking at thermal effects of intracanal erbium laser operation, George and Walsh measured temperature changes at the apical root surface during LAI with both Er:YAG and Er;Cr:YSGG (both in intracanal operation) under clinically relevant settings and found maximal increases of 2.5°C and concluded that both lasers can be safely used. Donnermeyer et al. measured the temperature of the irrigating solution at 2, 5 and 10 mm from the apex using thermocouples positioned through the canal wall under different irrigant activation regimens. They found that endoactivator and PIPS activation resulted in negligible temperature changes (Donnermeyer et al., 2021).

Penetration of lateral canals

Regarding penetration of lateral canals, Fidan and Erdemir created artificial lateral canals in roots of extracted monoradicular teeth and evaluated penetration of radiopaque irrigating solution activated with XP-endo Finisher, EDDY, Nd:YAG and Er:YAG laser. The highest lateral penetration values were observed in the Er:YAG LAI group (Fidan & Erdemir, 2023).

Smear layer removal

Many of the laboratory studies investigating the cleaning efficacy of LAI evaluated removal of smear layer from the root canal walls. Unfortunately, the huge majority of these studies suffer from substantial methodological deficiencies. It concerns studies in which SEM images of the root canal walls are evaluated using a score-based system. Limitations include the unknown status of the root canal wall prior to application of the irrigant (smear layer presence should be verified as a smear layer can be absent due to a lack of contact between instruments and canal wall), the nonrandomized and operator-dependent selection of the observation area (operator selection of the areas of SEM imaging instead of random selection can introduce bias), the subjective and qualitative evaluation of the SEM images, the lack of blinding and inter-examiner reproducibility verification, the lack of taking into account the amount and distribution of sclerotic dentin (dentinal sclerosis is a physiological aging effect, as a result tubuli are no longer visible on SEM), and the lack of a longitudinal character of the dentin surface observations (ideally each imaged root canal wall portion should be visualized both before and after the intervention) (De-Deus et al., 2011). Consequently, these kind of studies have been proven not to be trustworthy and reproducible (De-Deus et al., 2011). For these reasons, the smear layer removal studies falling into this category are not covered in this narrative view.

Dentin tubule penetration

The same holds true for studies evaluating irrigant penetration in dentinal tubules under a LAI regimen. Many

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such studies make use of dye penetration, but according to Boutsioukis et al. (2022) this setup is of limited value for various reasons: differences in physical properties between dye and irrigant, lack of a sharp interface marking the maximum penetration/clearance level complicating interpretation, the lack of taking into account the amount and distribution of dentinal sclerosis and the lack of using a two-step approach with initial dye penetration of patent tubules followed by NaOCl penetration in order to bleach the dye (Boutsioukis et al., 2022).

Laser safety and the dental operating microscope

As the eye is a structure that is vulnerable to laser injuries, Saegusa et al. (2010) evaluated the transmission of laser irradiation through a dental microscope. They found that there was zero transmission of Er:YAG radiation through the dental microscope. So the Er:YAG laser can be safely used under the dental operating microscope. However, it is advised that wavelength-matched eye protectors should always be worn by the patient and all involved personnel to protect the eyes from accidental exposures (Saegusa et al., 2010).

CONCLUSIONS

How pulsed erbium lasers deploy the power of cavitation to produce intense irrigant dynamics throughout the canal system is both fascinating and unique. Violently imploding cavitation bubbles around the laser tip cause different secondary phenomena in the irrigant, leading to intense liquid dynamics throughout the underlying pulpal space, and represents cleaning from a distance. Although troubled by significant methodological heterogeneity, studies in extracted teeth show LAI to result in superior canal disinfection than conventional irrigation, and there is a trend of higher antimicrobial efficacy of LAI over other activation methods. Clinical evidence is limited, with a number of trials demonstrating similar postoperative pain levels after LAI versus no activation or ultrasonic activation. Clinical evidence concerning the effect of LAI on healing of apical periodontitis is scarce, which underscores the need for future high-quality trials.

AUTHOR CONTRIBUTIONS

MM: conception, literature search, data analysis, critical appraisal, writing and reviewing. RDM: data analysis, critical appraisal and writing of the disinfection section.

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Maarten Meire reports nonfinancial support and Dr. Roeland De Moor reports personal fees from Fotona outside of the submitted work.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ETHIC STATEMENT

This investigation did not involve human or animal subjects.

PERMISSION TO REPRODUCE MATERIAL FROM OTHER SOURCES

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